

CSE4067

Introduction to Blockchain Programming

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08.10.2024

Bitcoin: Nodes

- Bitcoin itself consists of different users and nodes. We distinguish between **wallet owners**, **light nodes**, **full nodes** and **miners**.

Wallet Owner (User)

- The wallet owner owns different **private keys** to unspent transaction outputs (UTXOs).
- He is the **owner** of all stored currencies on these addresses.
- He sends money by **signing** and **publishing** new transactions to a connected light node, full node or miner.

Full Node (Software)

- The full node **maintains** the **complete blockchain**. Its record of the chain is complete, it contains every single transaction and block until the genesis block.
- Is **connected to** other **full nodes** and exchanges information. Namely:
 - **Validates** every transaction and block it receives
 - **Relays** all new transactions and blocks

Miner (Software)

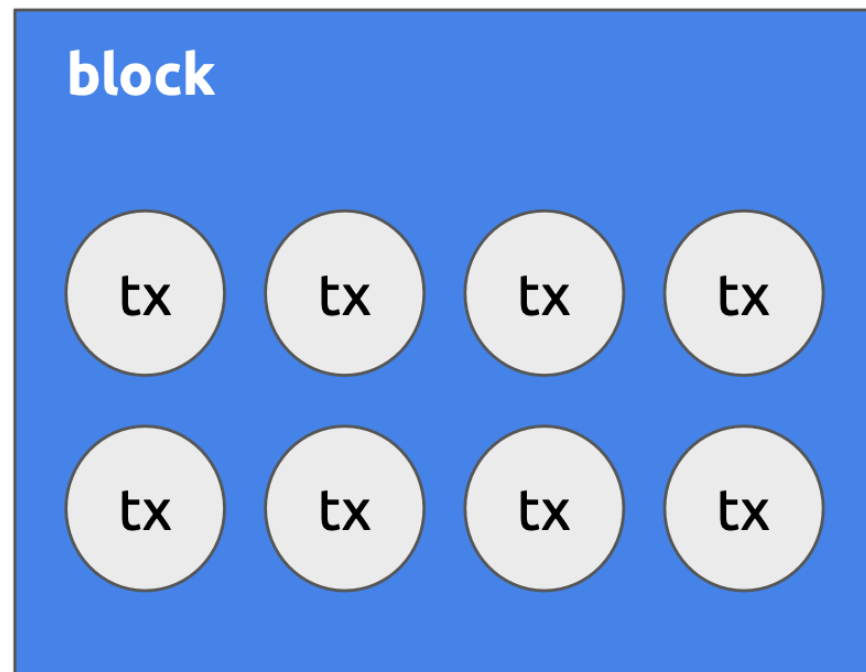
- The miner **needs** the **same record** as a **full node** in order to work properly. He also is connected to other nodes and maintains the network.
- Additionally, the miner is **responsible** for **creating** new **blocks** by trying to solve the **mining puzzle**.
- The miner gets rewarded by creating new blocks.

Light Node (Software)

- The light node can **act** as a **relay** for transactions of one **wallet owner**.
- It **validates** whether a **single transaction** of the wallet owner was executed correctly.
- The light node also **requires a full node** to connect to the network.
- **Almost no relevance** in practice today. Today, centralized services are used to create transactions.

Bitcoin: Block

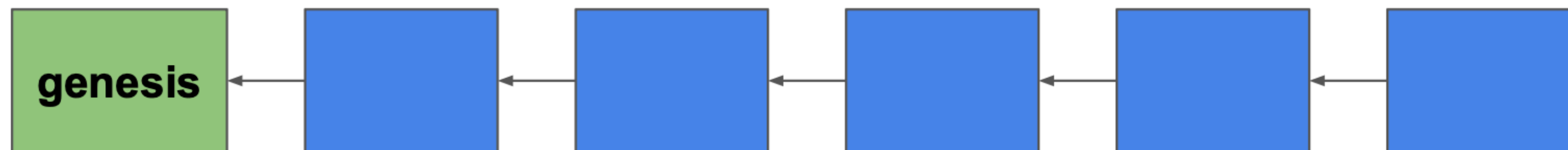
- Each block references a **previous** block
- This reference is by **hash** to its **previous** block



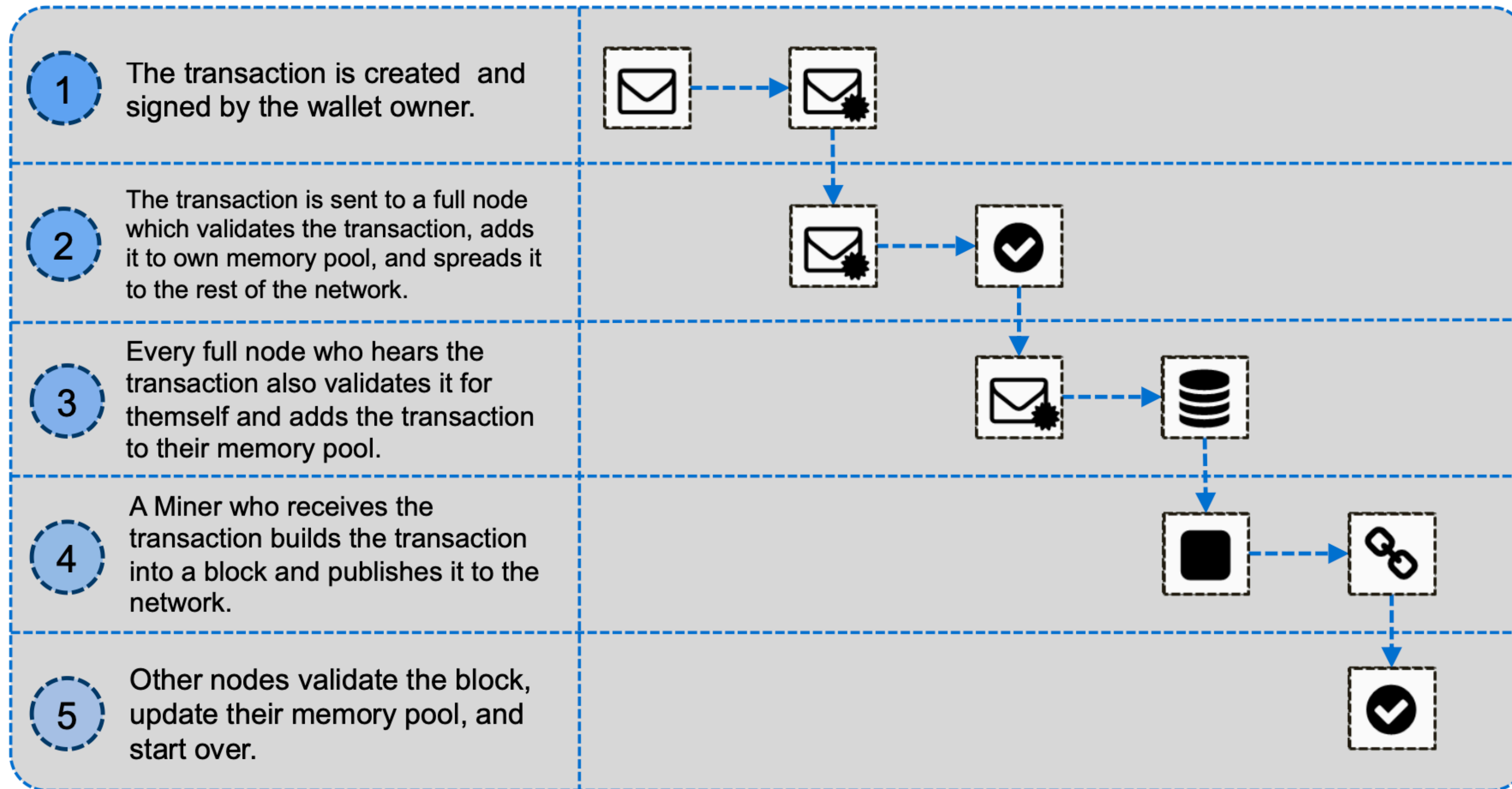
BLOCKS		TRANSACTIONS		
Height	Age	Transactions	Miner	Size (bytes)
564593	4 minutes	2734	Unknown	1,185,499
564592	9 minutes	2725	AntPool	1,297,232
564591	16 minutes	2537	BTC.com	1,183,625
564590	54 minutes	1757	F2Pool	1,158,256
564589	1 hour	2230	BitClub Network	1,300,144

Bitcoin: Genesis Block

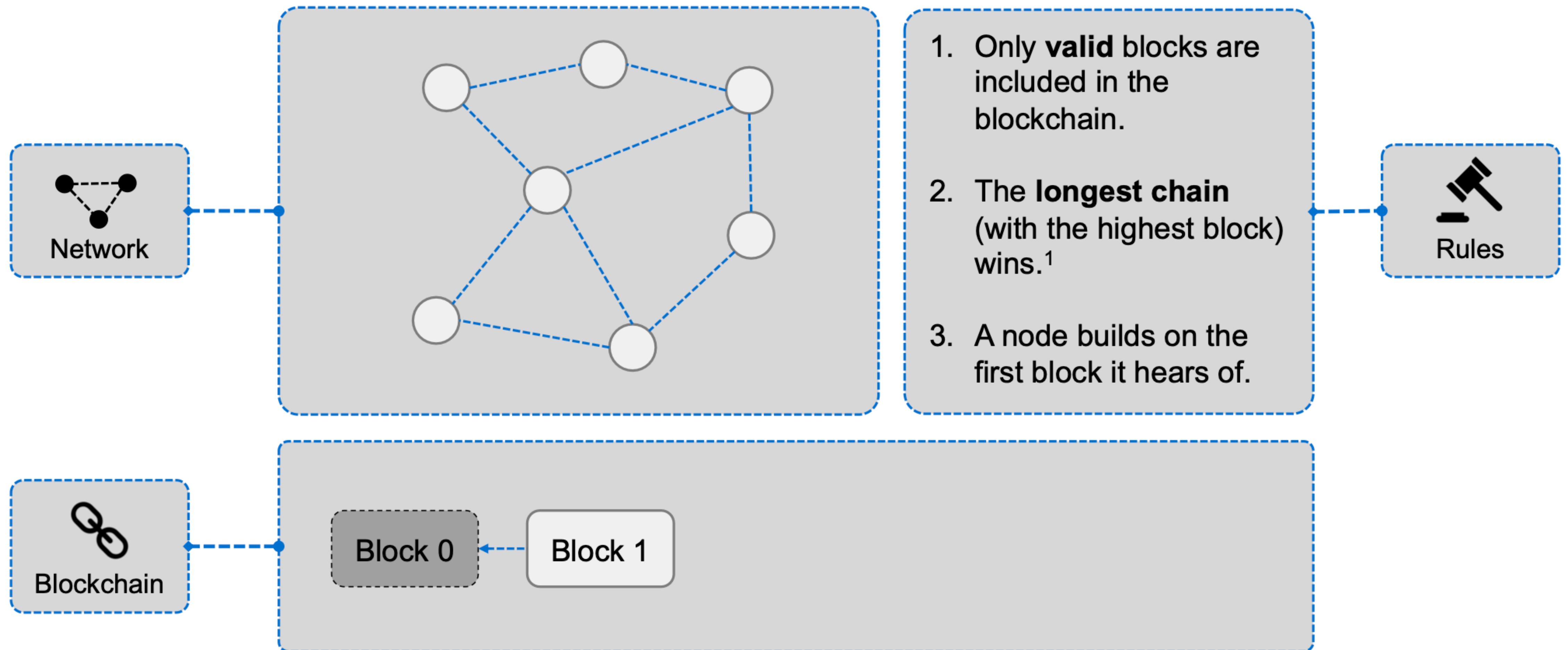
- Blocks have to reference their predecessor → **How does a blockchain start?**
- Bitcoin's genesis block:
 - mined at 2009-01-03 18:15:05
 - references a previous block with hash 0



Bitcoin: Transaction into Block

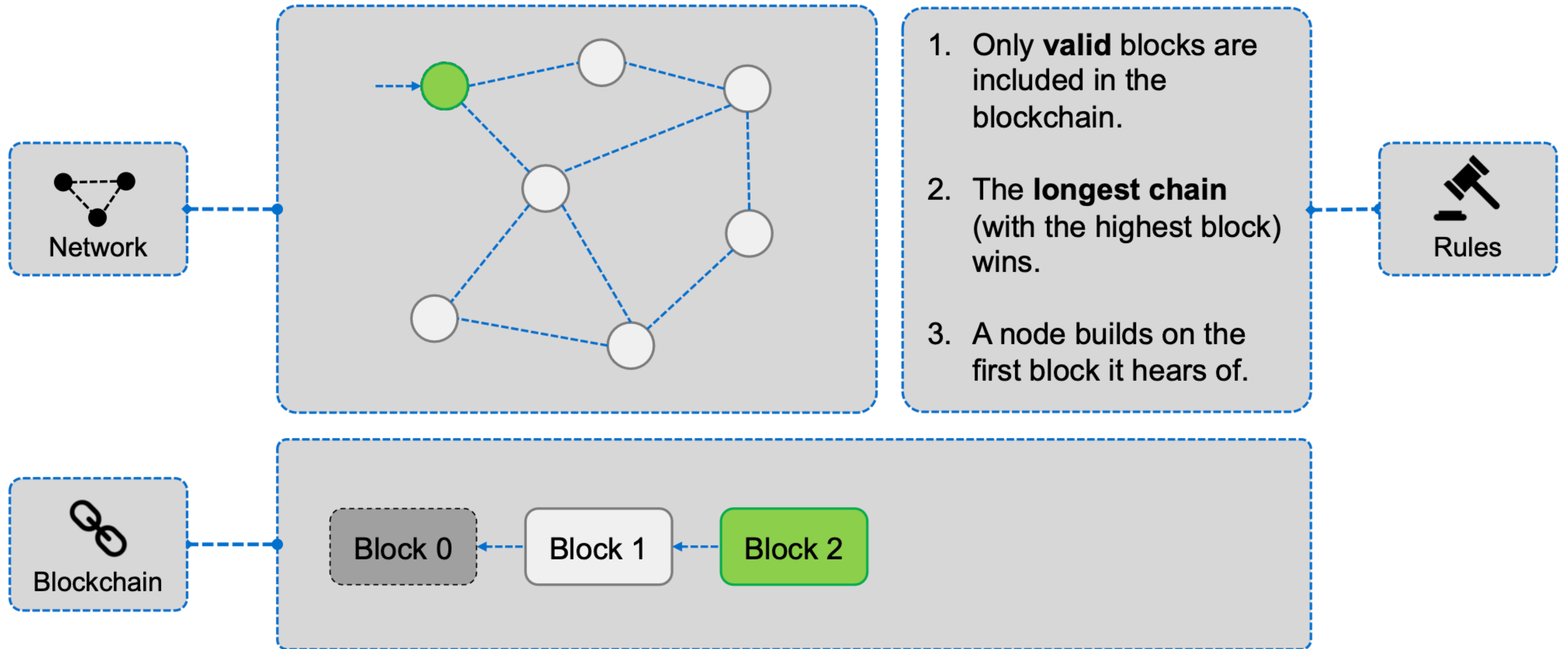


Block Propagation

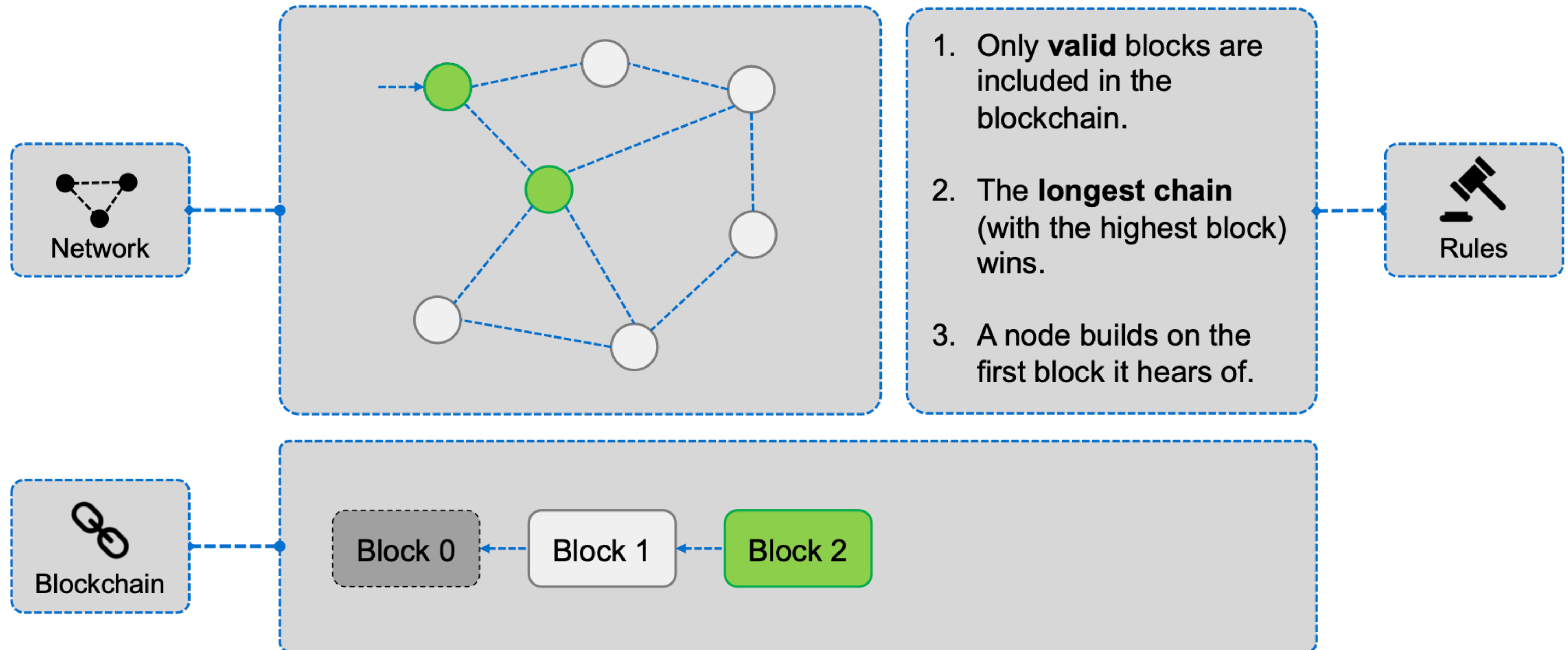


¹ The term "longest chain" refers to the chain with the highest weight in work done (i.e., the chain with the blocks that had the most difficult puzzles where miners submitted the most hashes while trying to solve them).

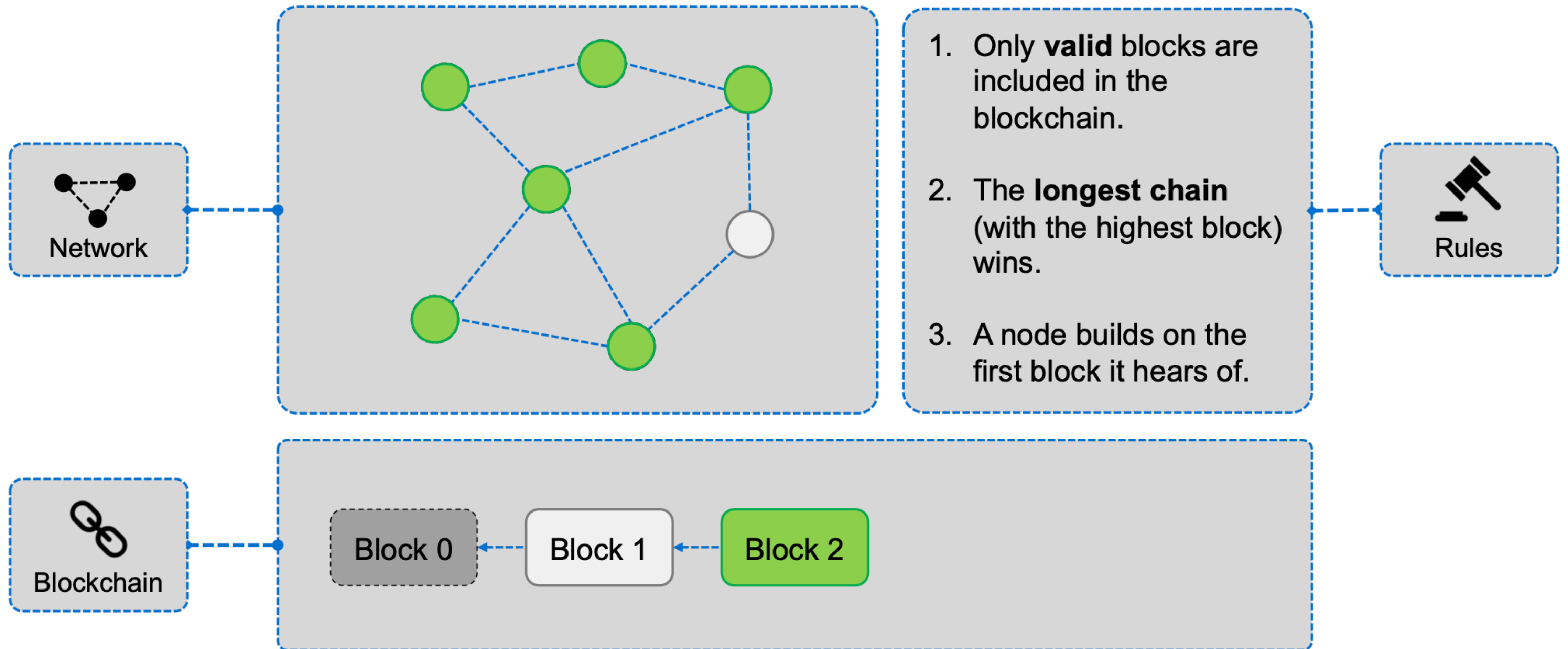
Block Propagation



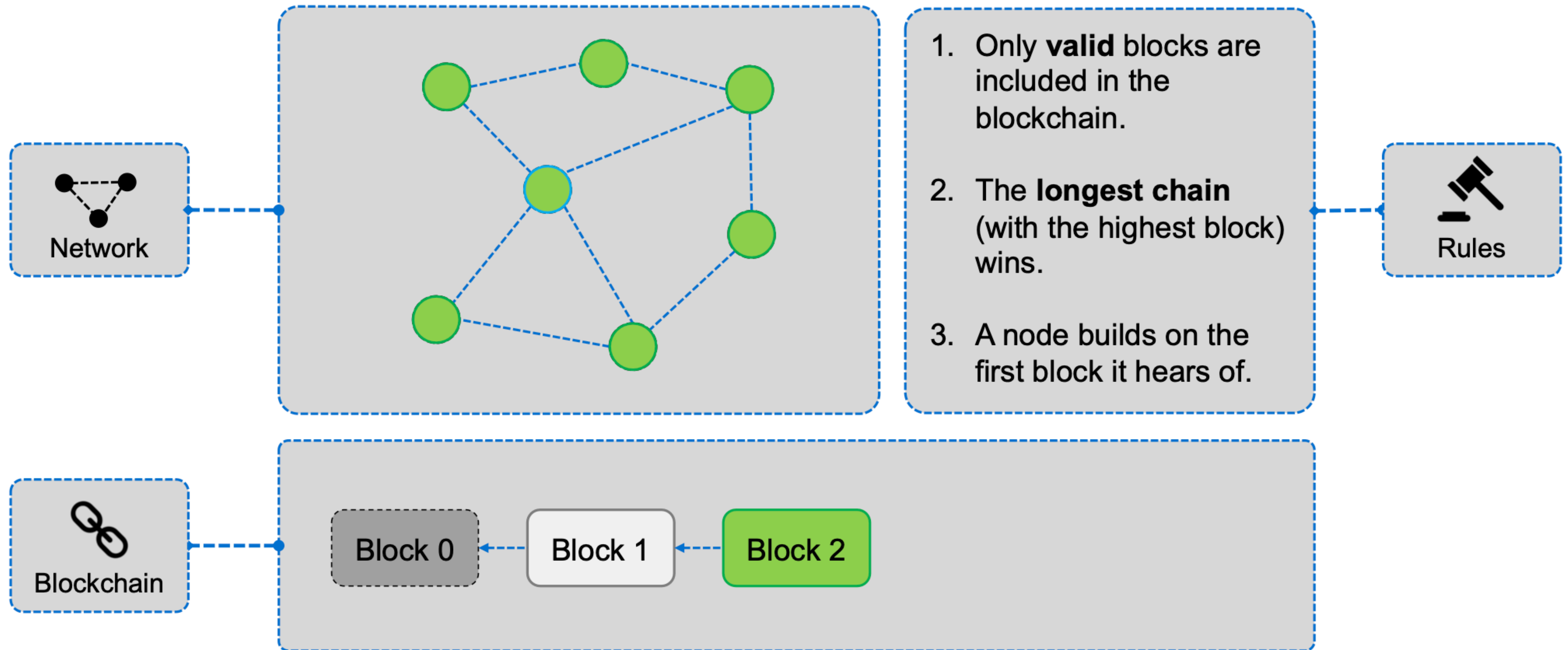
Block Propagation



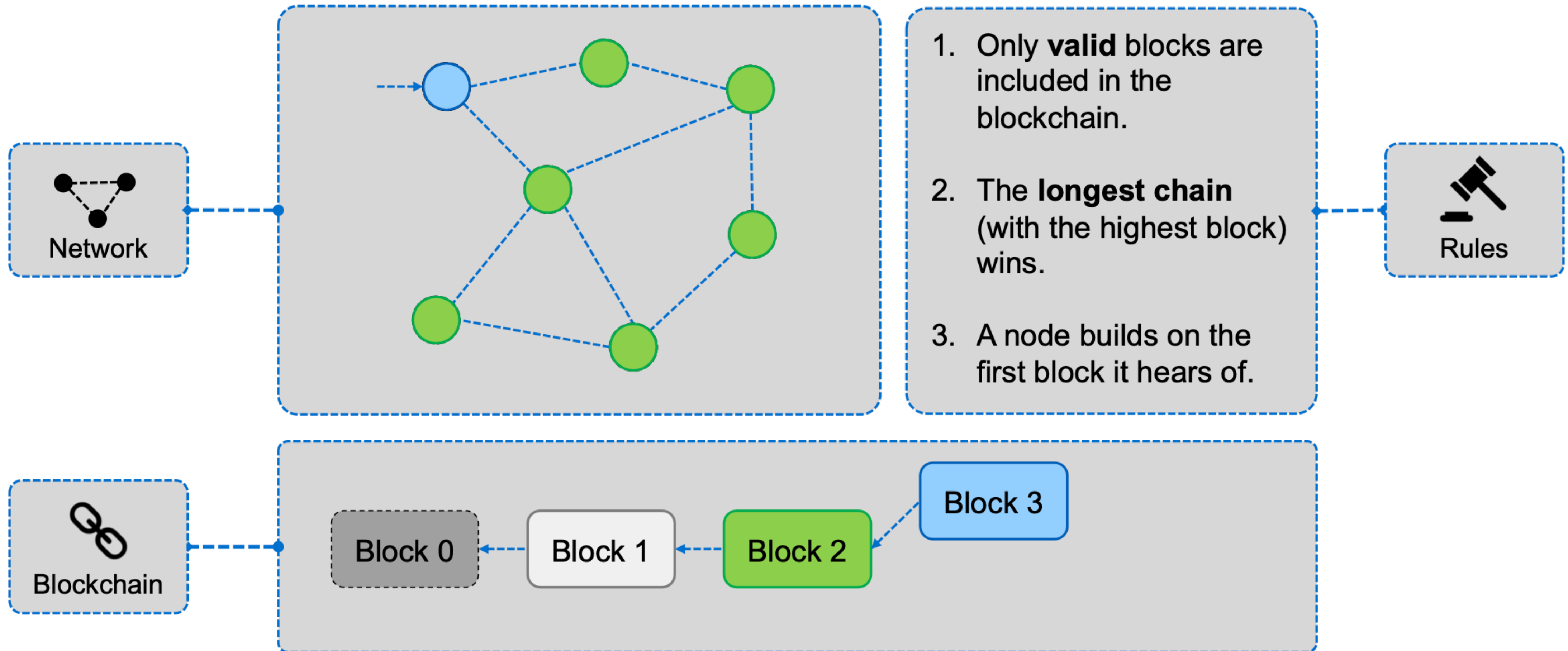
Block Propagation



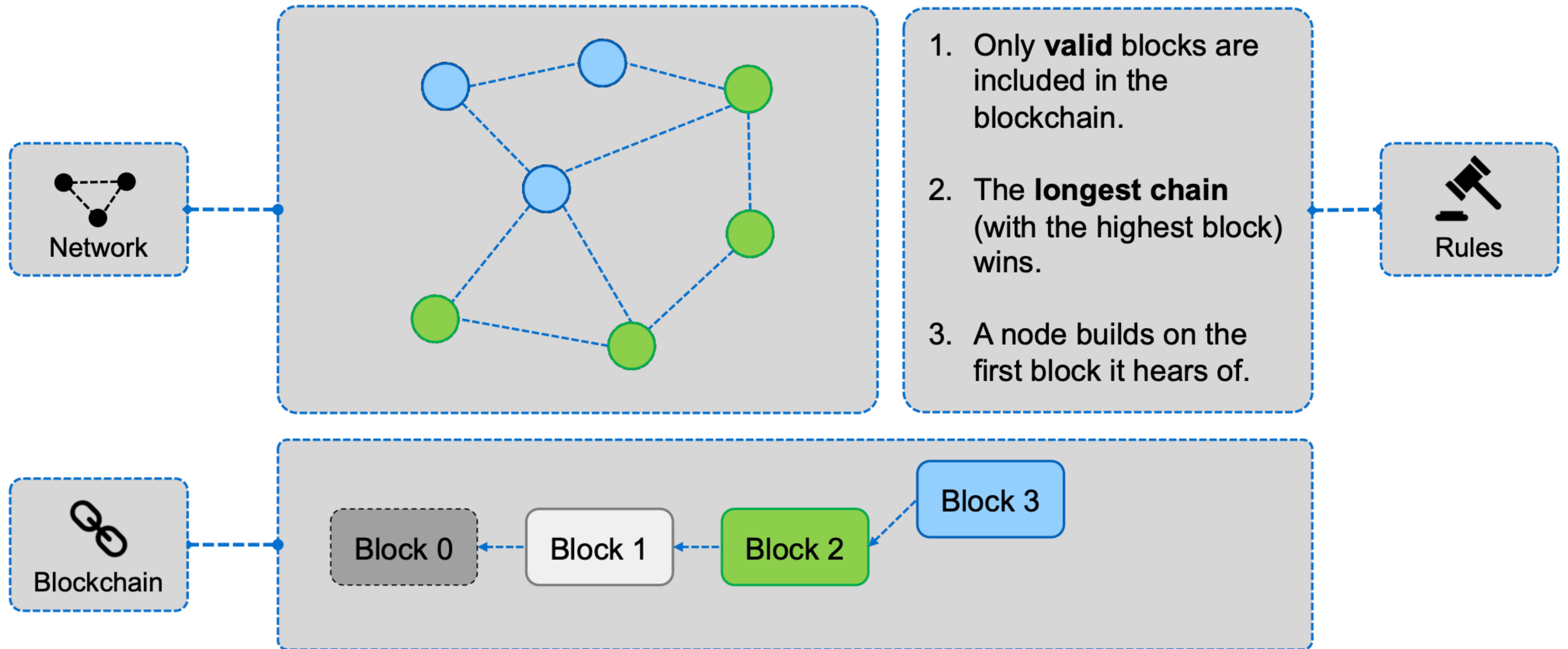
Block Propagation



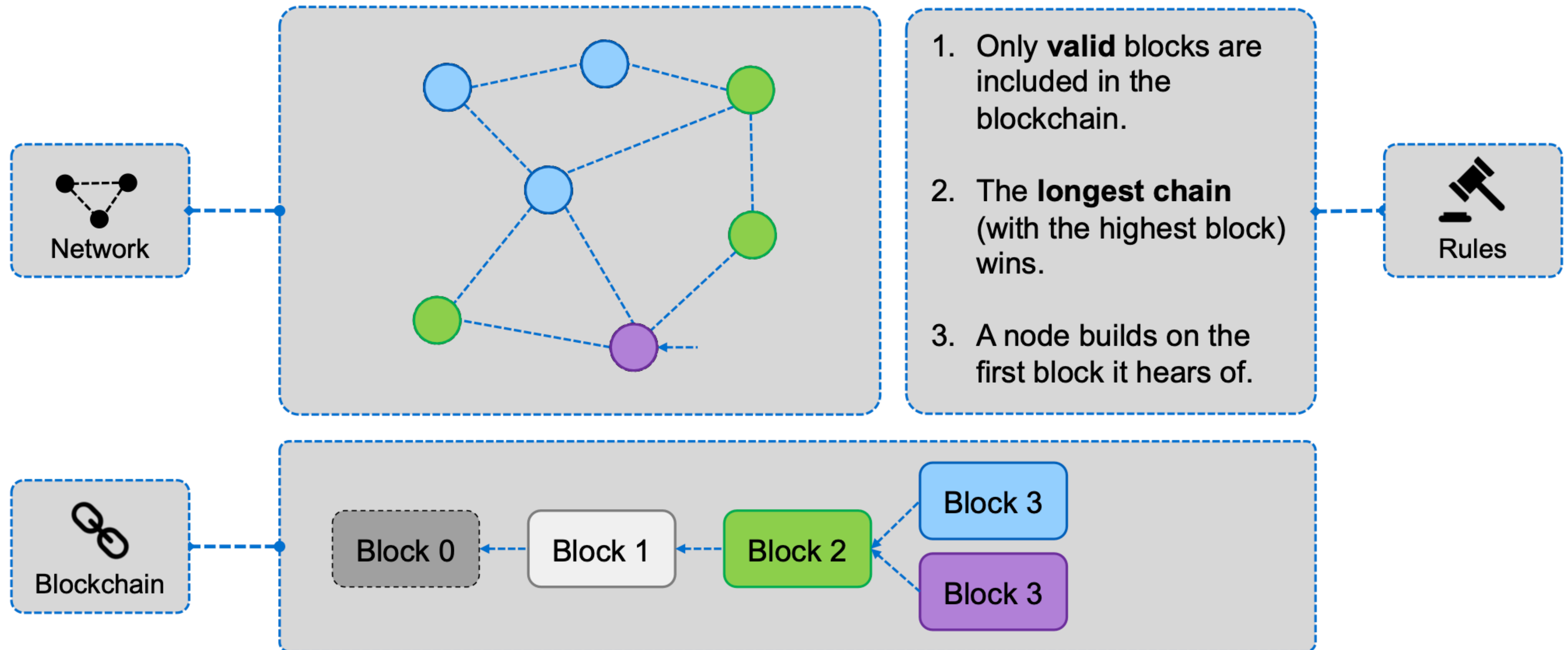
Block Propagation



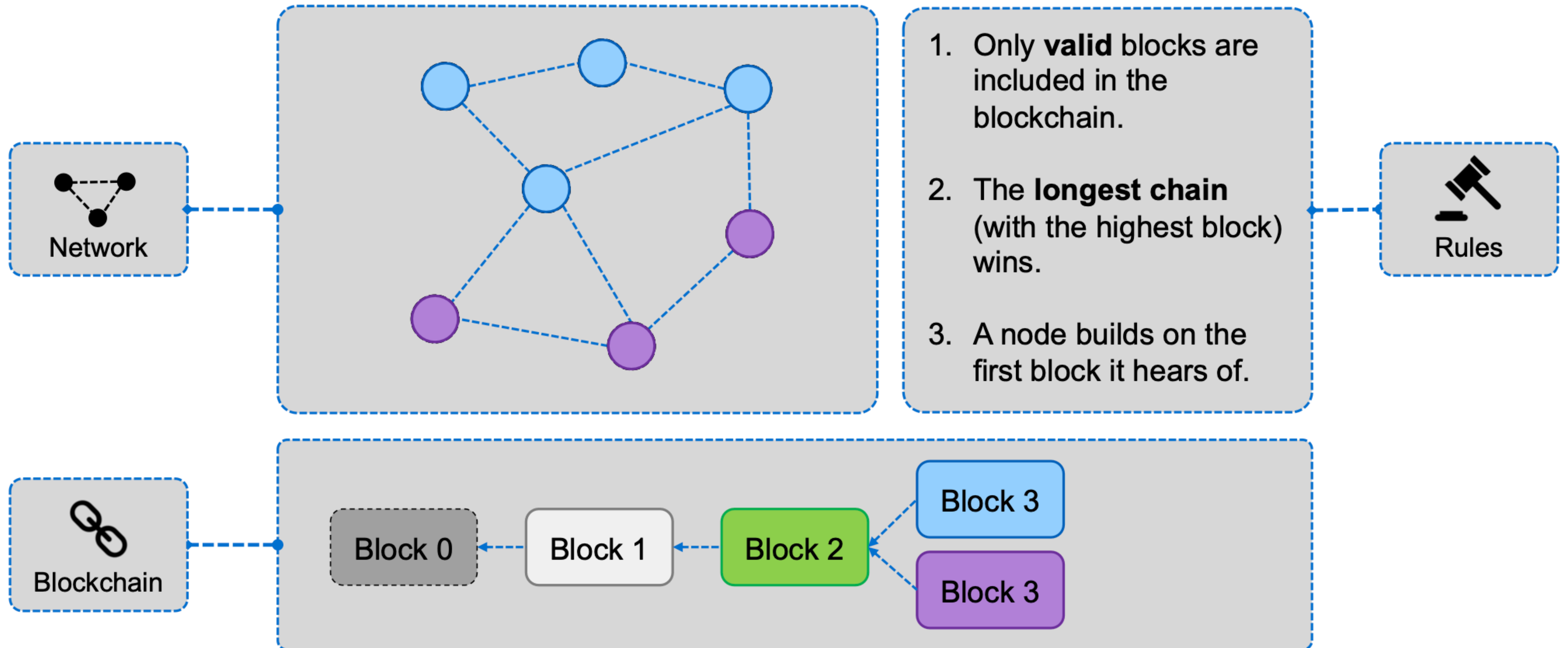
Block Propagation



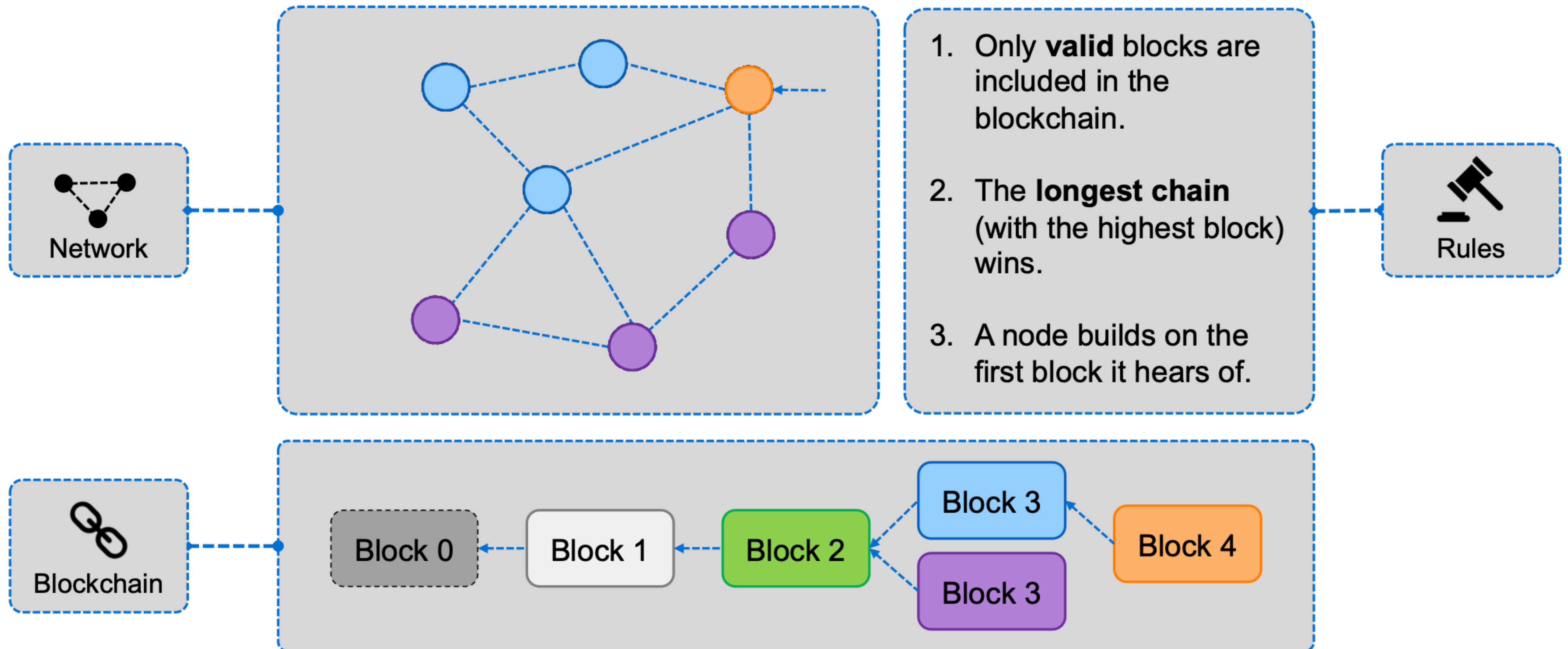
Block Propagation



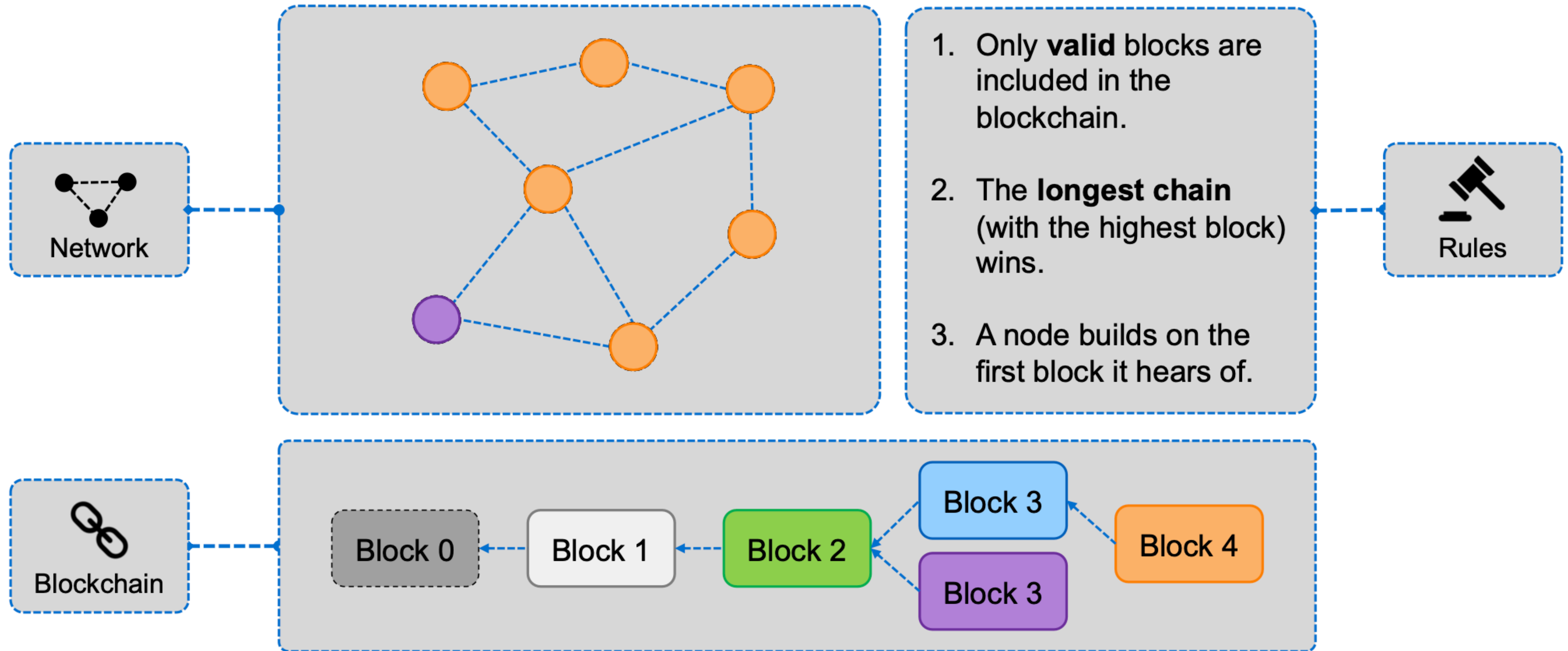
Block Propagation



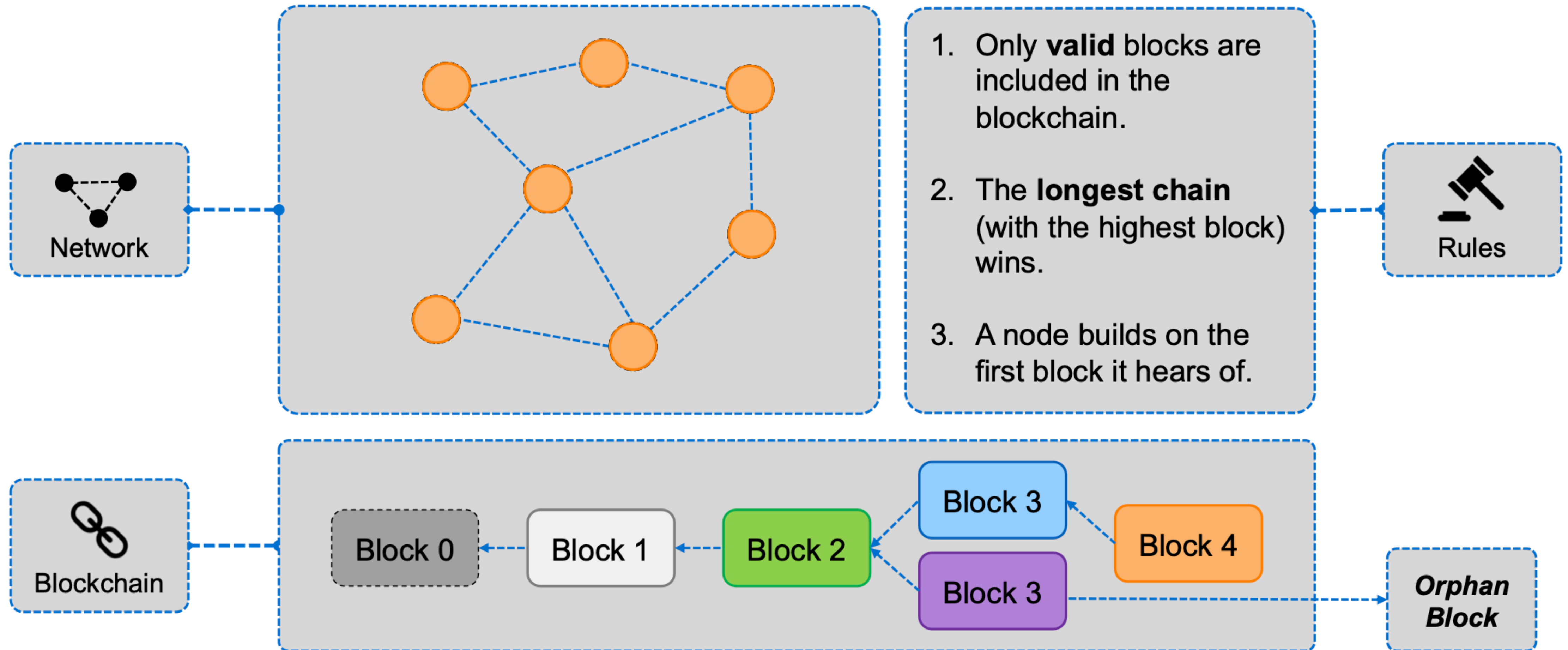
Block Propagation



Block Propagation



Block Propagation



Bitcoin: Wallets

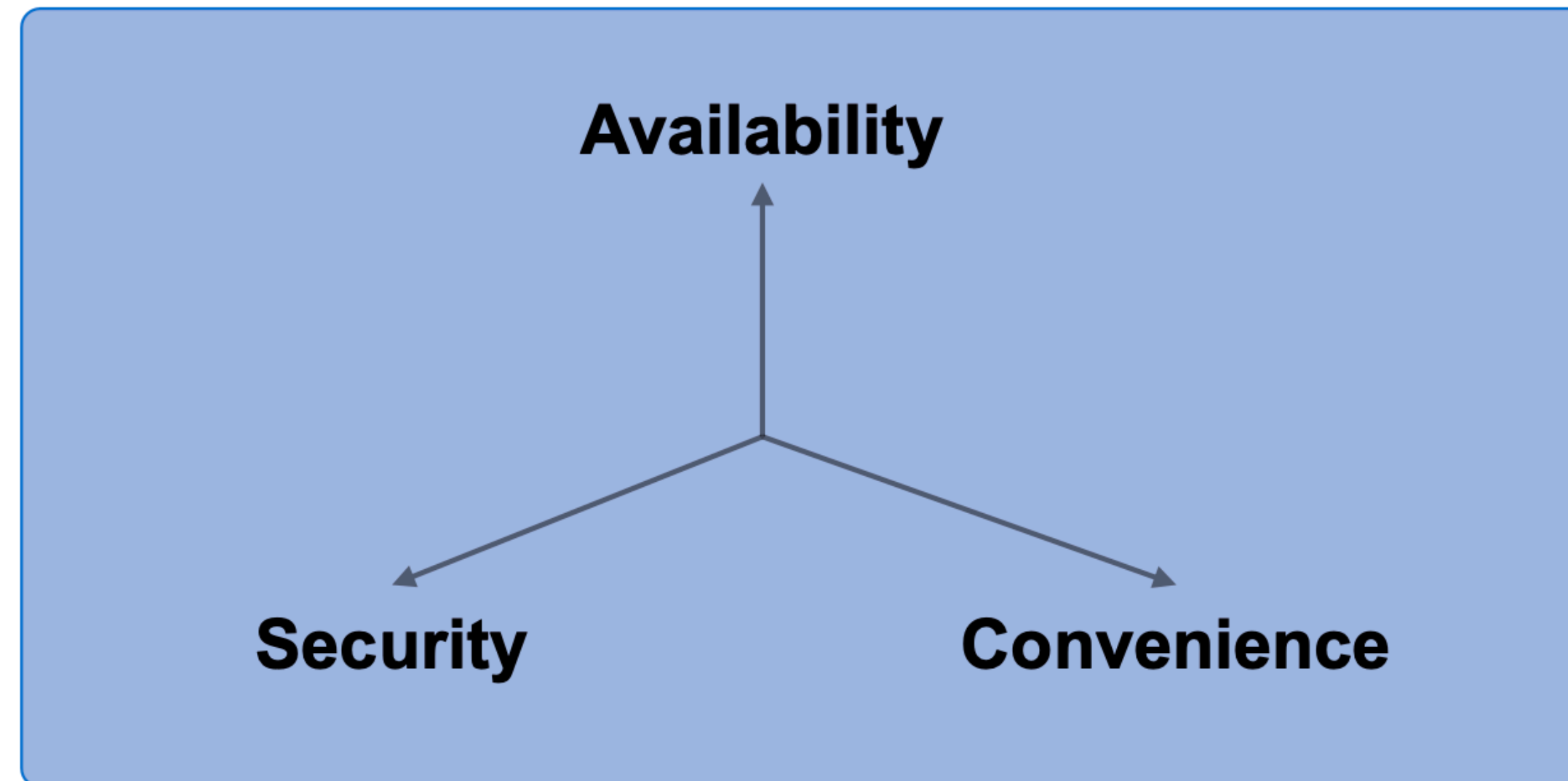
Storing Bitcoins is all about storing and managing secret keys.¹

Different approaches for storing and managing secret keys lead to different trade-offs between **availability**, **security** and **convenience**.

Availability: being able to access the keys when one wants to

Security: restricting access to the keys

Convenience: easy use



Bitcoin: Wallets

- Desktop Wallets
- Mobile Wallets (e.g. MetaMask)
 - **To-Do: Install MetaMask extension on Chrome and create a wallet!**
- Hardware Wallets



Bitcoin: Wallets

Cold Storage

- Takes some time to “activate”
- **Enhances security** at the cost of convenience and availability
- Advantage: Cold Storage does not have to be online to receive coins

Hot Storage

- **Is immediately available**
- Enables **convenience** and **availability** at the cost of security
- Example: Storage on your pc / mobile

Bitcoin: Wallets

Hardware Wallet

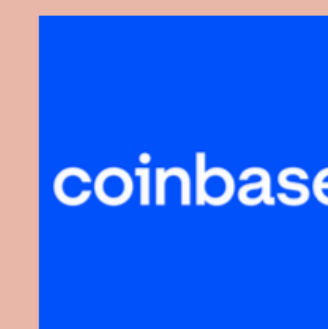
- **Key material** is stored on the **hardware device**.
- Additional: Generation of a 24-word passphrase. If device gets destroyed, the passphrase allows for a recovery.
- Device is designed to keep your private key private. Key is **securely stored** within the device.
- **The display shows the amount and target of a transaction.** The **keys** on the device allow for a **confirmation or rejection** of the transactions.
- Requires trust in manufacturer and intermediaries. (Never buy used hardware wallets!)



Hardware Wallet - Ledger Nano S

Online Wallet

- Let other people/ companies store your Bitcoins/ cryptocurrencies for you (custodial wallet).
- No access to the private key, coins can only be used through a certain interface / website.
- Very common within most exchanges. The money is sent to the exchange, the account on the platform has now a new balance which can be traded or paid out.
- However: **Very dangerous! Many exchanges got hacked, users lost their funds. Be careful!**



Bitcoin: Faucet

- A way to get test coins necessary for any testing.



- Ethereum (for **Sepolia** Testnet):
 - <https://cloud.google.com/application/web3/faucet/ethereum/sepolia>
 - **To-Do: Get test coins from the faucet to your MetaMask wallet. Find the faucet to work!**

Bitcoin: Explorers

- An online blockchain browser:
 - Displays the contents of individual blocks and transactions
 - Displays the transaction histories and balances of addresses.
 - Quick way to see if your transactions are confirmed.
- Bitcoin (for Mainnet):
 - <https://www.blockchain.com/explorer>
- Ethereum (for Mainnet):
 - <https://etherscan.io/>

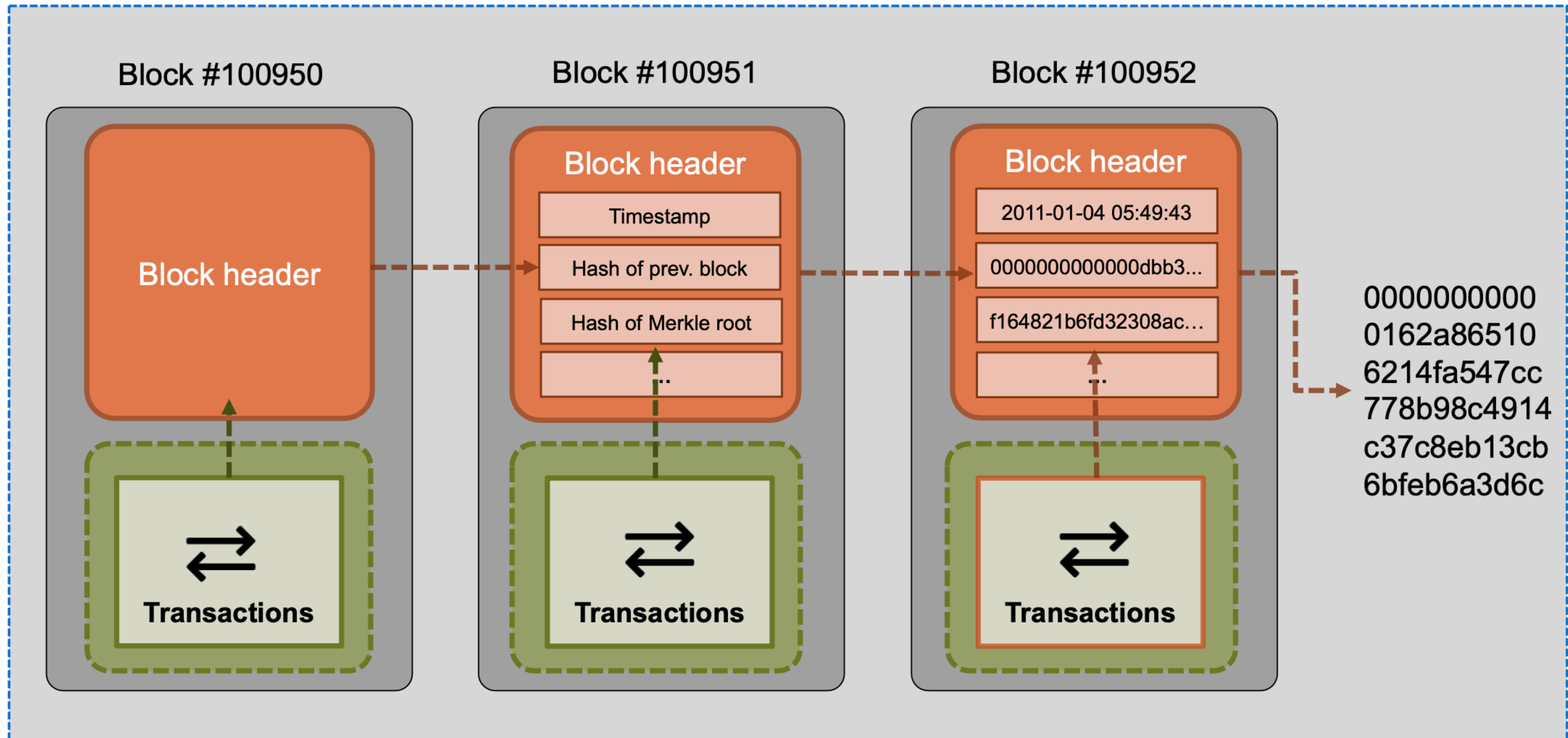
Bitcoin: Peer Discovery

- Each node stores a list of peers (by IP address)
- When Alice connects to Bob, Bob sends Alice his own known peers
- That way, Alice can learn about new peers
- Peer-to-peer nodes come “preinstalled” with some peers by IP / host
- When running a node, you can specify extra “known peers”

Bitcoin: Peer Discovery

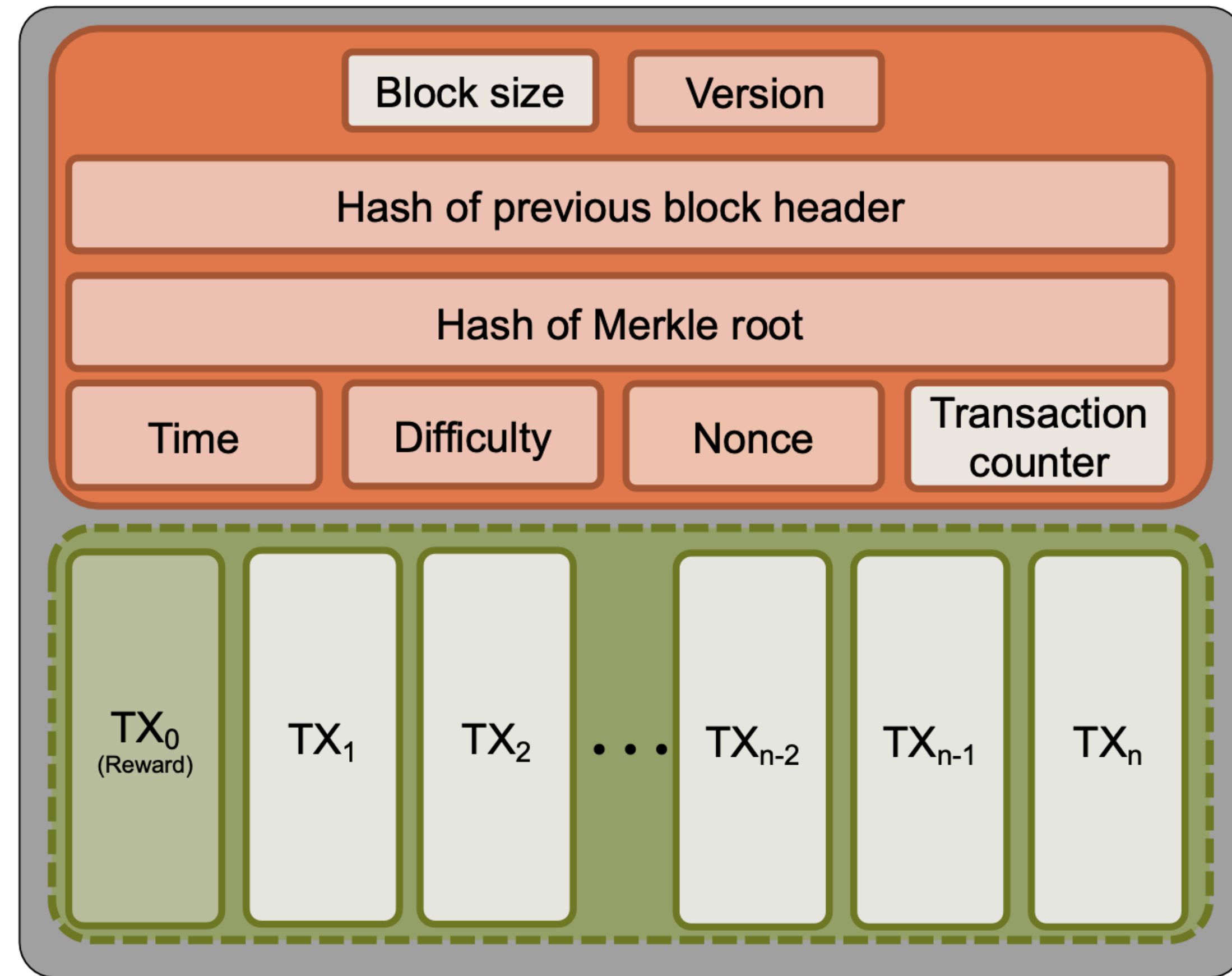
- **Gossip Protocol:**
 - When a node **Alice** generates some new data
 - Alice **broadcasts** data to its peers
 - Each peer broadcasts this data to *its* peers
 - If a peer has seen this data before, it ignores it
 - If this data is new, it broadcasts it to its peers
 - That way, the data spreads like an epidemic, until the whole network learns it
 - This process is called **diffuse**

Bitcoin: Blockchain



Bitcoin: Blockchain

- The hash of the previous block creates the chaining.
- The hash of the Merkle root node of a **Merkle tree** structure with all transactions.
- The **nonce** is required for the consensus mechanism in the network (used while solving the search puzzle).



Hash Function

- A **hash function** H accepts a variable-length block of data M as input and produces a fixed-size hash value $h = H(M)$.
- A **cryptographic hash function** is an algorithm for which it is computationally infeasible to find either
 - (a) a data object that maps to a pre-specified hash result (the one-way property) or
 - (b) two data objects that map to the same hash result (the collision-free property).

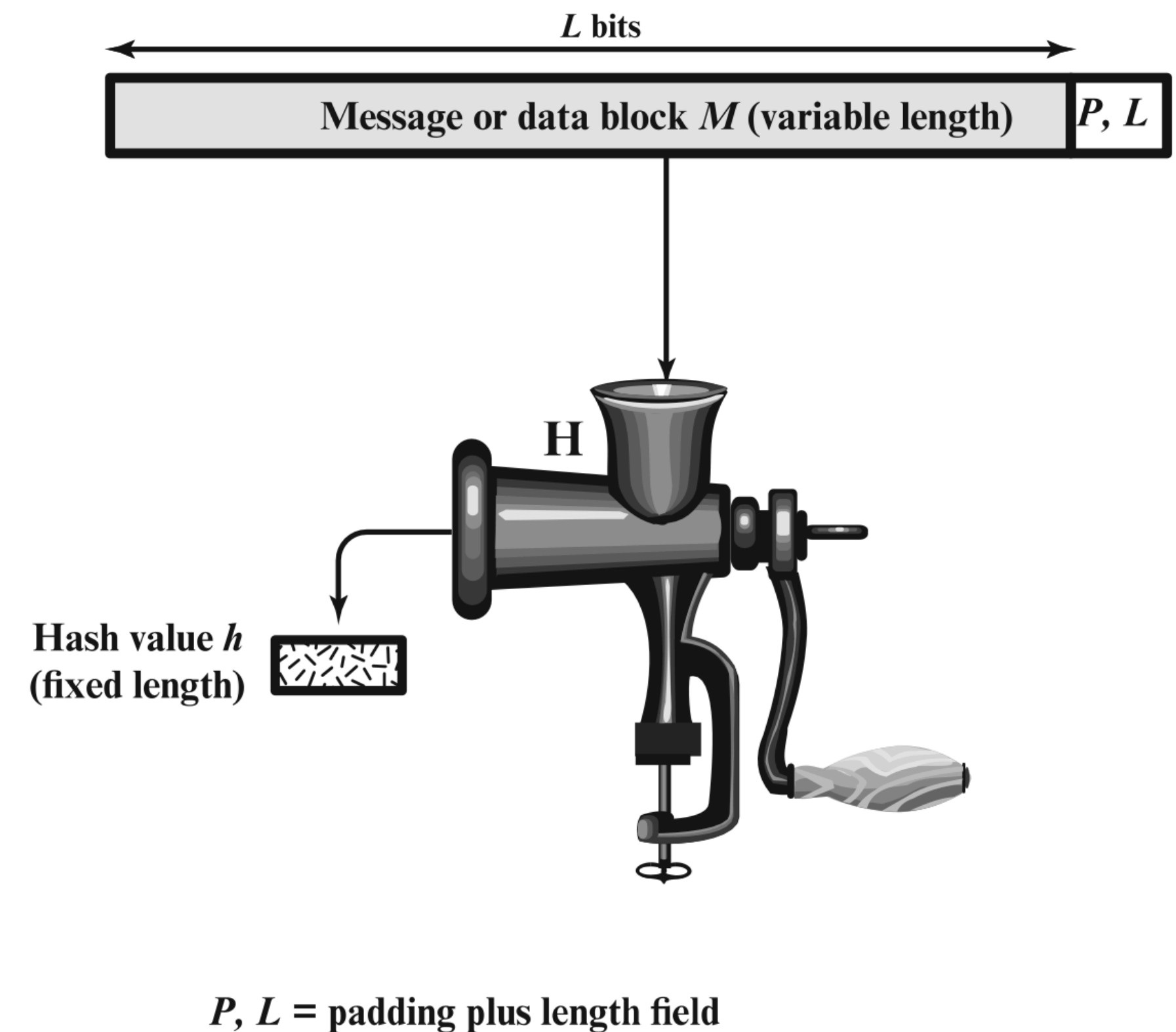


Figure 11.1 Cryptographic Hash Function; $h = H(M)$

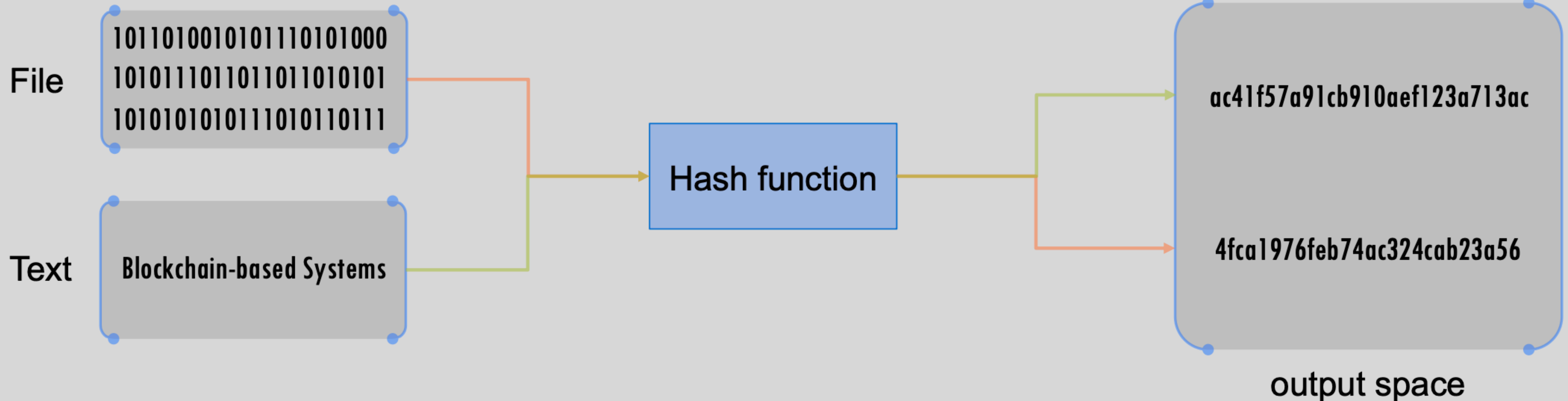
Hash Function

Definition: A function h is called a **hash function** if

- *Compression:* h maps an input x of arbitrary finite bit length to an output $h(x)$ of fixed bit length n :
 $h: \{0,1\}^* \rightarrow \{0,1\}^n$
- *Ease of computation:* Given h and x it is easy to compute $h(x)$

Compression

Independent from the size of the input, the output of h is within a fixed space.



Requirements of Hash Function

Requirement	Description
Variable input size	H can be applied to a block of data of any size.
Fixed output size	H produces a fixed-length output.
Efficiency	$H(x)$ is relatively easy to compute for any given x , making both hardware and software implementations practical.
Preimage resistant (one-way property)	For any given hash value h , it is computationally infeasible to find y such that $H(y) = h$.
Second preimage resistant (weak collision resistant)	For any given block x , it is computationally infeasible to find $y \neq x$ with $H(y) = H(x)$.
Collision resistant (strong collision resistant)	It is computationally infeasible to find any pair (x, y) with $x \neq y$, such that $H(x) = H(y)$.
Pseudorandomness	Output of H meets standard tests for pseudorandomness.